Supplementary Material

Grain-Boundary Environments Reorient the Local $4f$ Susceptibility Axis and Split Light from Heavy Lanthanides in $Nd_2Fe_{14}B$

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S1 Structures and embedding definitions

Tables S1 and S2 give every atom and the lattice vectors of the relaxed 68-atom bulk and 136-atom interface cells. The accompanying `bulk_CONTCAR` and `GB_CONTCAR` contain the same structures. Table S3 lists the rare-earth coordination. The upper half of the interface cell obeys $(x,y,z) \mapsto (-y,x,z+1/2)$ modulo lattice translations, without an additional in-plane translation. The 90° rotation is symmetry equivalent at the tetragonal orientation level. The model is a coherent rotational-registry interface, not a representative general high-angle boundary.

The finite-cell calculations in the main figures use bulk atom 2 ($4g$) and interface atom 13 (class A, $4f$ parent). The remaining atoms of one cell carry rare-earth $+3$, Fe 0, and B -3 charges. There are 67 and 135 entries, with charge sums $+9e$ and $+21e$, respectively. The quantum ion has charge $+3e$; it is not a point-charge entry. A point-charge sum is therefore a property of the embedding, and is not by itself a validation or invalidation of its local potential. The finite-cell prescription does not reproduce a charge-neutral periodic Madelung potential by construction.

The extended-sphere sensitivity model uses periodic images of the same structures within 18 or 25 Å of the ion, with rare-earth $+3$, Fe $-3/14$, and B -3. The residual embedding charge is spread equally over the charges in the outermost 2 Å shell. For example, the 25 Å arrays for Nd at bulk atom 2 and interface atom 13 contain 4768 and 4760 entries; the printed arrays are neutral to their rounding precision. This model changes the Fe charge assignment as well as the truncation geometry and size. Only the comparison of 18 and 25 Å within this prescription is a radius-sensitivity test. An alternative neutral partition uses Fe $-5/14$ and B -1. The matched pairs are A (atom 13) versus bulk $4f$ (atom 5), and B (atom 9) versus bulk $4g$ (atom 2). Finite-cell and sphere-model results are labeled separately throughout.

S2 Orbital and magnetic descriptors

The active spaces, state-average roots and weights, relativistic basis and SOC treatment are specified in the main Methods. The finite-cell calculations use SA-CASSCF; the separate SC-NEVPT2 results are labeled in Table S6. The one-electron AILFT energies ε_i yield the total orbital spread and the orbital gaps $\delta_{\mathrm{orb}} = \varepsilon_2 - \varepsilon_1$ and $\Delta_{\mathrm{orb}} = \varepsilon_3 - \varepsilon_2$. Magnetic excitations instead come from the many-electron QDPT-SOC eigenvalues (Table S4). For Pr and Tb, the table distinguishes the splitting within the lowest pair from the next level; a quasi-doublet need not be a protected degeneracy. The half-filled Gd ground multiplet is weakly split, despite a substantial one-electron orbital spread.

The AILFT matrix is decomposed exactly as $V = \sum_{kq} B^k_q C^k_q$ for $k = 2,4,6$, using the angular-momentum matrices printed by ORCA to fix the real-f basis convention. The coefficients satisfy $B^k_{-q} = (-1)^q \overline{B^k_q}$. We report $S_k = [\sum_q |B^k_q|^2/(2k+1)]^{1/2}$ and $A^0_2\langle r^2\rangle = B^2_0/2$ in the cell frame with $z \parallel c$. Table S5 supplies the finite-cell invariants and axial coefficient; `wybourne_Bkq_all_runs.csv` supplies the complete complex coefficient sets for the archived calculations, including the additional embedding tests. This CSV is a separate supplementary data file, not an embedded table in the PDF. Figure S2 compares rank changes for the finite-cell and matched sphere-model comparisons. Rank decomposition is not a ligand-shell decomposition or a radial-density measurement.

The susceptibility figure uses the tensor-trace response at zero static field, orientationally averaged as $\overline{\chi}_M T = T\,\mathrm{Tr}(\chi_M)/3$. Molar normalization is per rare-earth ion. The main Fig. 3 directions are the eigen-

vectors of the largest eigenvalues of the directly computed zero-field tensors at 2 K. The Gd susceptibility-axis rotation is 87.08° and is included in main Fig. 3; this is not an assignment of first-order orbital anisotropy. The susceptibility axes are evaluated directly from the saved output tensors. Only records with the explicit `_bulk` and `_GB` labels define the pair; auxiliary `_casscf` records are excluded. The archived direction table is based on the principal direction of the printed `g(OZ/SOC)` contribution, not the full ground-doublet g tensor. With the correct records, that direction indicator gives 87.01°, 87.34°, 87.67°, 87.26°, and 87.42° for Ce, Pr, Nd, Tb, and Dy. The direct 2 K susceptibility directions used in main Fig. 3 differ by less than 0.04° from these indicators; the two definitions are not equated. The susceptibility tensor defines the actual linear magnetic response at the stated temperature for both Kramers and non-Kramers ions, whereas the archived indicator uses only one contribution to an effective g matrix. Figure 4 uses the trace of each printed zero-field $\chi_M T$ tensor, from the archived outputs; the separate scalar `d2E/dB2` column is not substituted for that trace. The orbital maps in main Fig. 5 use the archived orbital-27 cubes for bulk and interface Ce, Pr and Dy. The panel titles give the ion, environment (Bulk or GB), and MO27 file index; the axes have labeled 0.5 Å intervals and 0.25 Å minor ticks. The projected density is $P(a,b) = \int |\psi(a,b,c)|^2 dc/N$, where $N = \int |\psi|^2 d^3r$ is the numerical full-cube norm. All six panels use the common range $0 \leq P \leq P_{\max}$, with $P_{\max} = 7$ Å^{-2} and the square-root color coordinate $u = (P/P_{\max})^{1/2}$. The colorbar is labeled in physical P values (Å^{-2}), not in u. The same monotonic mapping is applied to every panel to make weak lobes visible; there is no independent panel normalization, density clipping, or change to the integrals and radii. This is a display transformation, not a replacement of the density by an orbital amplitude. Thin black contours outline the projected-density lobes at 0.25, 0.75, and 2.0 Å^{-2} in every panel. They are isodensity contours, not coordination boundaries. Bicubic interpolation is used only to display the contours; the cube integrals and radii use the original three-dimensional grid. The extent is $r_{\mathrm{rms}} = [\int r^2 |\psi|^2 d^3r/N]^{1/2}$, measured from the ion using all three-dimensional voxels. The uncorrected grid norms differ from one by less than 0.9% (Table S10); each density is normalized with its own N before comparison. Projecting the full density avoids an arbitrary slice through an orbital nodal plane. The selected individual eigenorbital is not the many-electron ground-state density, and its angular form can change under mixing of nearly degenerate orbitals. Its extent does not by itself establish a causal radial mechanism for the crystal-field crossover.

S3 Embedding and site sensitivity

Figure S1 and Tables S6 and S7 distinguish finite-cell versus sphere-model changes, the radius test within the sphere model, and matched-site comparisons. The selected-site finite-cell sign crossover is not retained in the matched 25 Å sphere calculations. The sphere model yields a class A increase of the axial term and a smaller class B response, but this does not demonstrate that this charge assignment is uniquely correct. At the tested Nd and Dy sites, changing sphere radius modifies axial parameters by about 5–8%; changing Fe/B charge partition can modify them by 33–78%. The 13–14% change of the ratio of interface atom 13 to bulk atom 2 is not an uncertainty bound on the different matched-sublattice ratios.

The cECP tests in Table S8 compare core-repulsion potentials with bare charges at the same def2-TZVP/central-ion-ECP level, using spin-free SA-CASSCF. Every Fe and rare-earth neighbor within 4.0 Å carries its formal charge and, in the capped arm, an embedding ECP but no basis functions. The capped and bare-charge arms differ only by these potentials. Their small differences do not test explicit Fe–4f hybridization, conduction-electron screening, or alternative interface structures. They also do not validate the separate finite-cell charge convention.

S4 Active-space checks and magnetic directions

The Ce 4f-parent sphere calculations did not retain seven f active orbitals: one active orbital acquired 6s character. A CASCI on free-ion orbitals did not agree adequately with SA-CASSCF where both were available,

so the missing Ce class A entries are omitted. This is an active-space limitation, not a calculation of Ce valence in the solid.

For the sphere-model Kramers ions Ce, Nd and Dy, the direction in Table S7 is taken directly from the ground-doublet matrix elements of $\mathbf{L} + g_e\mathbf{S}$. Directions for the non-Kramers ions Pr and Tb are not assigned in that table because direct temperature-dependent susceptibility tensors were not saved for these sphere calculations. No Gd orbital easy axis is assigned. The main finite-cell directions instead use the saved direct susceptibility tensors for all six ions, with Gd reported as a susceptibility response direction. No fitted ligand-field CI reconstruction is used for the main susceptibility curves, magnetic directions, or orbital-density figure.

S5 Conditional switching-field estimate

For a single uniaxial coherent-rotation region, $E = K\sin^2\phi - \mu_0 M_s H\cos(\phi - \theta)$ yields the Stoner–Wohlfarth loss-of-metastability field $H_{\mathrm{sw}}/H_A = (|\cos\theta|^{2/3} + |\sin\theta|^{2/3})^{-3/2}$, with $H_A = 2K/(\mu_0 M_s)$. Table S9 evaluates this dimensionless expression for the finite-cell susceptibility-axis angles, assuming the applied field is along the bulk response axis. The result is not the hysteretic coercivity along a hard direction, and it is not a spatially resolved nucleation calculation. It omits exchange between the region and host, demagnetizing fields, changes of K or M_s, and nonuniform reversal. No value of K is inferred from the orbital-energy spread, and no absolute coercive field is assigned.

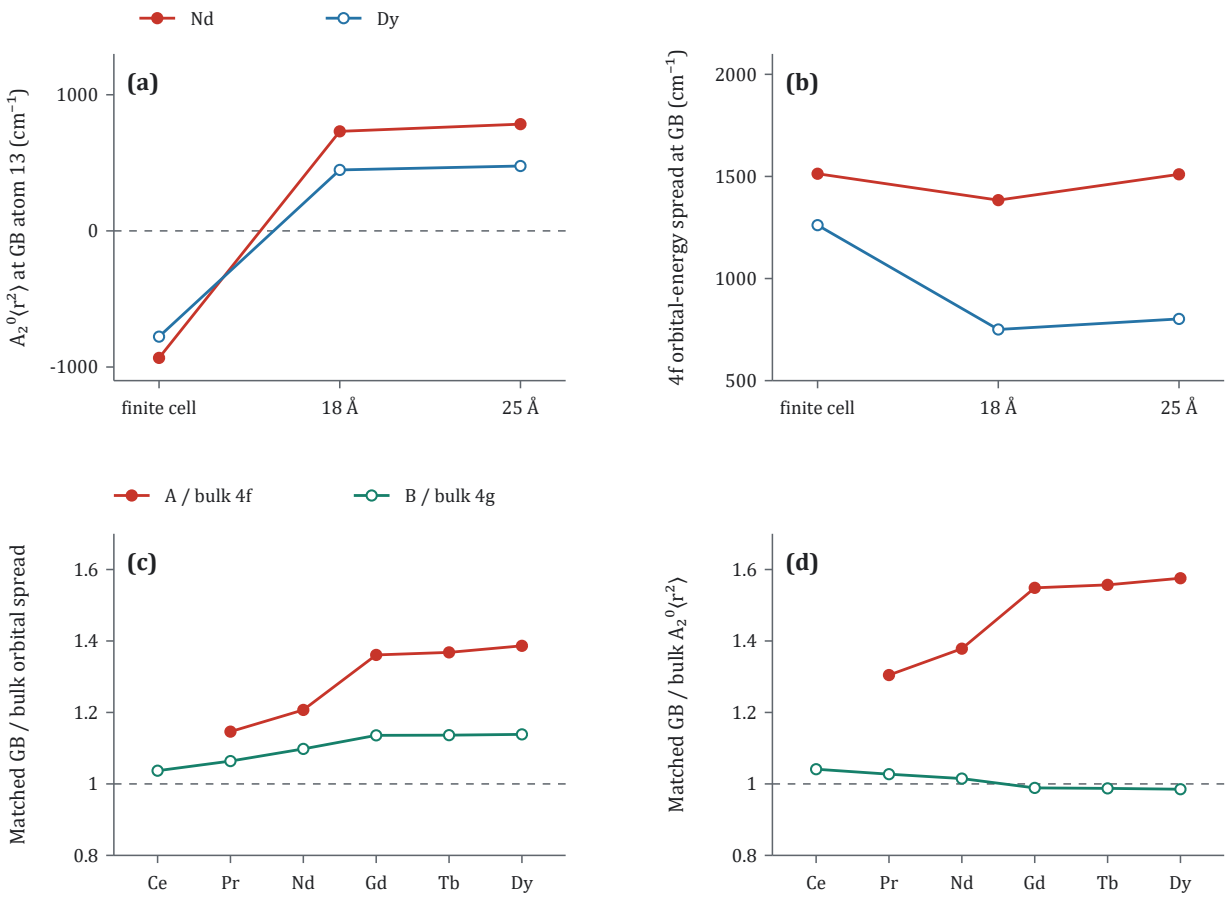


Figure S1: Embedding and site comparisons. (a) Axial parameter at interface atom 13 and (b) one-electron orbital-energy spread for Nd and Dy in the finite-cell model and the 18 and 25 Å sphere models. The first step changes charge assignment as well as truncation; the second step tests radius within one charge convention. (c) Orbital-spread ratios and (d) axial-parameter ratios for matched pairs in the 25 Å sphere model. The Ce class A pair is absent because the active-space optimization was unstable. Lines guide the eye.

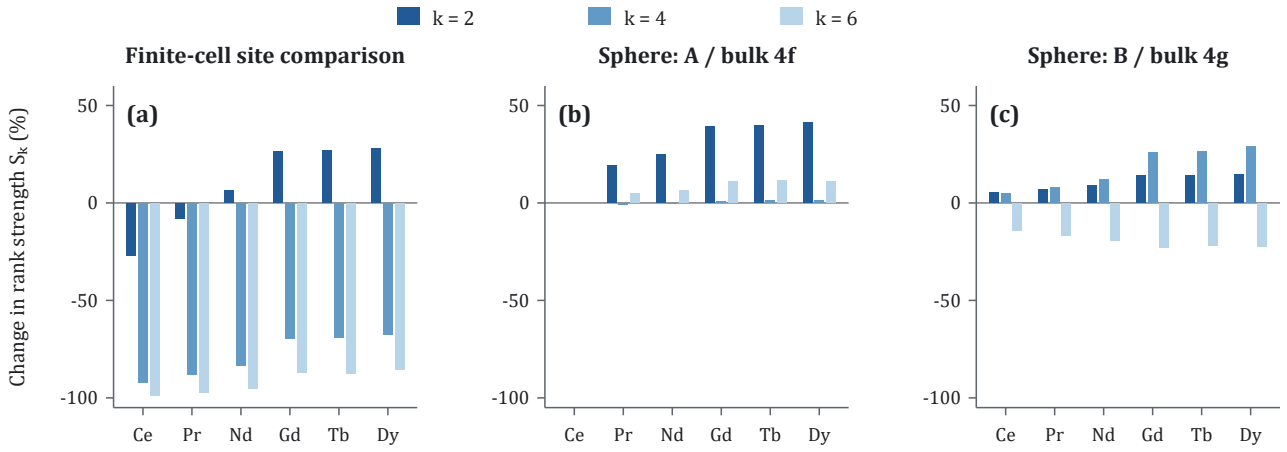


Figure S2: Relative changes of exact rank invariants S_k ($k = 2,4,6$). (a) Finite-cell selected-site comparison, bulk atom 2 versus interface atom 13; (b) matched class A versus bulk $4f$ and (c) matched class B versus bulk $4g$ in the 25 Å sphere model. The different patterns show that the finite-cell sign crossover cannot be treated as an embedding-independent geometric effect. The decomposition does not isolate radial contraction.

Table S1: Relaxed bulk $Nd_8Fe_{56}B_4$ supercell (68 atoms, space group $P4_2/mnm$). Lattice vectors (Å): a_1 = (8.756856, 0.000000, 0.000000), a_2 = (-0.000000, 8.756856, 0.000000), a_3 = (-0.000000, -0.000000, 12.129949). Fractional (x, y, z) and Cartesian $(X, Y, Z$, Å) coordinates; atom indices are those used in the main text.

#	El.	x	y	z	X	Y	Z	Site
1	Nd	0.85705	0.14295	-0.00000	7.5051	1.2518	0.0000	4g
2	Nd	0.35705	0.35705	0.50000	3.1267	3.1267	6.0650	4g
3	Nd	0.64295	0.64295	0.50000	5.6302	5.6302	6.0650	4g
4	Nd	0.14295	0.85705	-0.00000	1.2518	7.5051	0.0000	4g
5	Nd	0.73144	0.73144	-0.00000	6.4051	6.4051	0.0000	4f
6	Nd	0.76856	0.23144	0.50000	6.7302	2.0267	6.0650	4f
7	Nd	0.23144	0.76856	0.50000	2.0267	6.7302	6.0650	4f
8	Nd	0.26856	0.26856	-0.00000	2.3517	2.3517	0.0000	4f
9	Fe	0.56740	0.22461	0.12756	4.9687	1.9669	1.5473	
10	Fe	0.43260	0.77539	0.12756	3.7882	6.7899	1.5473	
11	Fe	0.22461	0.56740	0.87244	1.9669	4.9687	10.5826	
12	Fe	0.72461	0.93260	0.62756	6.3453	8.1666	7.6123	
13	Fe	0.27539	0.06740	0.62756	2.4115	0.5902	7.6123	
14	Fe	0.93260	0.72461	0.37244	8.1666	6.3453	4.5176	
15	Fe	0.06740	0.27539	0.37244	0.5902	2.4115	4.5176	
16	Fe	0.77539	0.43260	0.87244	6.7899	3.7882	10.5826	
17	Fe	0.43260	0.77539	0.87244	3.7882	6.7899	10.5826	
18	Fe	0.56740	0.22461	0.87244	4.9687	1.9669	10.5826	
19	Fe	0.77539	0.43260	0.12756	6.7899	3.7882	1.5473	
20	Fe	0.27539	0.06740	0.37244	2.4115	0.5902	4.5176	
21	Fe	0.72461	0.93260	0.37244	6.3453	8.1666	4.5176	
22	Fe	0.06740	0.27539	0.62756	0.5902	2.4115	7.6123	
23	Fe	0.93260	0.72461	0.62756	8.1666	6.3453	7.6123	
24	Fe	0.22461	0.56740	0.12756	1.9669	4.9687	1.5473	
25	Fe	0.35996	0.03683	0.17626	3.1521	0.3225	2.1381	
26	Fe	0.64004	0.96317	0.17626	5.6047	8.4343	2.1381	
27	Fe	0.03683	0.35996	0.82374	0.3225	3.1521	9.9919	
28	Fe	0.53683	0.14004	0.67626	4.7010	1.2263	8.2031	
29	Fe	0.46317	0.85996	0.67626	4.0559	7.5306	8.2031	
30	Fe	0.14004	0.53683	0.32374	1.2263	4.7010	3.9269	
31	Fe	0.85996	0.46317	0.32374	7.5306	4.0559	3.9269	
32	Fe	0.96317	0.64004	0.82374	8.4343	5.6047	9.9919	
33	Fe	0.64004	0.96317	0.82374	5.6047	8.4343	9.9919	
34	Fe	0.35996	0.03683	0.82374	3.1521	0.3225	9.9919	
35	Fe	0.96317	0.64004	0.17626	8.4343	5.6047	2.1381	
36	Fe	0.46317	0.85996	0.32374	4.0559	7.5306	3.9269	
37	Fe	0.53683	0.14004	0.32374	4.7010	1.2263	3.9269	
38	Fe	0.85996	0.46317	0.67626	7.5306	4.0559	8.2031	
39	Fe	0.14004	0.53683	0.67626	1.2263	4.7010	8.2031	
40	Fe	0.03683	0.35996	0.17626	0.3225	3.1521	2.1381	
41	Fe	0.90221	0.90221	0.79526	7.9006	7.9006	9.6465	
42	Fe	0.09779	0.09779	0.79526	0.8563	0.8563	9.6465	
43	Fe	0.90221	0.90221	0.20474	7.9006	7.9006	2.4835	
44	Fe	0.40221	0.59779	0.29526	3.5221	5.2347	3.5815	

#	El.	x	y	z	X	Y	Z	Site
45	Fe	0.59779	0.40221	0.29526	5.2347	3.5221	3.5815	
46	Fe	0.59779	0.40221	0.70474	5.2347	3.5221	8.5485	
47	Fe	0.40221	0.59779	0.70474	3.5221	5.2347	8.5485	
48	Fe	0.09779	0.09779	0.20474	0.8563	0.8563	2.4835	
49	Fe	0.68217	0.68217	0.75430	5.9737	5.9737	9.1496	
50	Fe	0.31783	0.31783	0.75430	2.7832	2.7832	9.1496	
51	Fe	0.68217	0.68217	0.24570	5.9737	5.9737	2.9804	
52	Fe	0.18217	0.81783	0.25430	1.5952	7.1616	3.0846	
53	Fe	0.81783	0.18217	0.25430	7.1616	1.5952	3.0846	
54	Fe	0.81783	0.18217	0.74570	7.1616	1.5952	9.0453	
55	Fe	0.18217	0.81783	0.74570	1.5952	7.1616	9.0453	
56	Fe	0.31783	0.31783	0.24570	2.7832	2.7832	2.9804	
57	Fe	-0.00000	0.00000	0.61492	0.0000	0.0000	7.4589	
58	Fe	0.50000	0.50000	0.11492	4.3784	4.3784	1.3939	
59	Fe	0.50000	0.50000	0.88508	4.3784	4.3784	10.7360	
60	Fe	-0.00000	0.00000	0.38508	0.0000	0.0000	4.6710	
61	Fe	0.50000	0.00000	-0.00000	4.3784	0.0000	0.0000	
62	Fe	0.50000	0.00000	0.50000	4.3784	0.0000	6.0650	
63	Fe	-0.00000	0.50000	0.50000	-0.0000	4.3784	6.0650	
64	Fe	-0.00000	0.50000	-0.00000	-0.0000	4.3784	0.0000	
65	B	0.62335	0.37665	-0.00000	5.4586	3.2982	0.0000	
66	B	0.12335	0.12335	0.50000	1.0802	1.0802	6.0650	
67	B	0.87665	0.87665	0.50000	7.6767	7.6767	6.0650	
68	B	0.37665	0.62335	-0.00000	3.2982	5.4586	0.0000	

Table S2: Relaxed 90° [001] twist grain-boundary supercell $Nd_{16}Fe_{112}B_8$ (136 atoms). Interface planes at $z = 0$ and $z = 1/2$; the rotation specifies the coherent registry construction (Section S1). Lattice vectors (Å): a_1 = (8.776004, 0.000000, 0.000000), a_2 = (-0.000000, 8.776004, 0.000000), a_3 = (0.000000, 0.000000, 24.238864). Fractional (x, y, z) and Cartesian $(X, Y, Z$, Å) coordinates; atom indices are those used in the main text.

#	El.	x	y	z	X	Y	Z	Site
1	Nd	0.84544	0.15456	0.00216	7.4196	1.3564	0.0523	class B
2	Nd	0.35610	0.35610	0.25023	3.1251	3.1251	6.0653	class C
3	Nd	0.64390	0.64390	0.25023	5.6509	5.6509	6.0653	class C
4	Nd	0.15456	0.84544	0.00216	1.3564	7.4196	0.0523	class B
5	Nd	0.75274	0.75274	-0.00146	6.6061	6.6061	-0.0355	class A
6	Nd	0.76445	0.23555	0.25005	6.7088	2.0672	6.0609	class D
7	Nd	0.23555	0.76445	0.25005	2.0672	6.7088	6.0609	class D
8	Nd	0.24726	0.24726	-0.00146	2.1699	2.1699	-0.0355	class A
9	Nd	0.84544	0.84544	0.50216	7.4196	7.4196	12.1717	class B
10	Nd	0.64390	0.35610	0.75023	5.6509	3.1251	18.1847	class C
11	Nd	0.35610	0.64390	0.75023	3.1251	5.6509	18.1847	class C
12	Nd	0.15456	0.15456	0.50216	1.3564	1.3564	12.1717	class B
13	Nd	0.24726	0.75274	0.49854	2.1699	6.6061	12.0839	class A
14	Nd	0.76445	0.76445	0.75005	6.7088	6.7088	18.1803	class D
15	Nd	0.23555	0.23555	0.75005	2.0672	2.0672	18.1803	class D

#	El.	x	y	z	X	Y	Z	Site
16	Nd	0.75274	0.24726	0.49854	6.6061	2.1699	12.0839	class A
17	Fe	0.55952	0.22634	0.06334	4.9104	1.9864	1.5353	
18	Fe	0.44048	0.77366	0.06334	3.8656	6.7897	1.5353	
19	Fe	0.22916	0.47739	0.44344	2.0111	4.1895	10.7484	
20	Fe	0.72310	0.93917	0.31070	6.3459	8.2422	7.5310	
21	Fe	0.27690	0.06083	0.31070	2.4301	0.5338	7.5310	
22	Fe	0.93460	0.72426	0.18595	8.2021	6.3562	4.5071	
23	Fe	0.06540	0.27574	0.18595	0.5739	2.4199	4.5071	
24	Fe	0.77084	0.52261	0.44344	6.7649	4.5865	10.7484	
25	Fe	0.52261	0.77084	0.44344	4.5865	6.7649	10.7484	
26	Fe	0.47739	0.22916	0.44344	4.1895	2.0111	10.7484	
27	Fe	0.77366	0.44048	0.06334	6.7897	3.8656	1.5353	
28	Fe	0.27574	0.06540	0.18595	2.4199	0.5739	4.5071	
29	Fe	0.72426	0.93460	0.18595	6.3562	8.2021	4.5071	
30	Fe	0.06083	0.27690	0.31070	0.5338	2.4301	7.5310	
31	Fe	0.93917	0.72310	0.31070	8.2422	6.3459	7.5310	
32	Fe	0.22634	0.55952	0.06334	1.9864	4.9104	1.5353	
33	Fe	0.35905	0.03426	0.08832	3.1510	0.3007	2.1407	
34	Fe	0.64095	0.96574	0.08832	5.6250	8.4753	2.1407	
35	Fe	0.01341	0.35651	0.41034	0.1177	3.1287	9.9463	
36	Fe	0.53025	0.14444	0.33880	4.6535	1.2676	8.2121	
37	Fe	0.46975	0.85556	0.33880	4.1225	7.5084	8.2121	
38	Fe	0.14078	0.53525	0.16169	1.2355	4.6974	3.9192	
39	Fe	0.85922	0.46475	0.16169	7.5405	4.0786	3.9192	
40	Fe	0.98659	0.64349	0.41034	8.6583	5.6473	9.9463	
41	Fe	0.64349	0.98659	0.41034	5.6473	8.6583	9.9463	
42	Fe	0.35651	0.01341	0.41034	3.1287	0.1177	9.9463	
43	Fe	0.96574	0.64095	0.08832	8.4753	5.6250	2.1407	
44	Fe	0.46475	0.85922	0.16169	4.0786	7.5405	3.9192	
45	Fe	0.53525	0.14078	0.16169	4.6974	1.2355	3.9192	
46	Fe	0.85556	0.46975	0.33880	7.5084	4.1225	8.2121	
47	Fe	0.14444	0.53025	0.33880	1.2676	4.6535	8.2121	
48	Fe	0.03426	0.35905	0.08832	0.3007	3.1510	2.1407	
49	Fe	0.90461	0.90461	0.38996	7.9388	7.9388	9.4522	
50	Fe	0.09539	0.09539	0.38996	0.8372	0.8372	9.4522	
51	Fe	0.90319	0.90319	0.10257	7.9264	7.9264	2.4862	
52	Fe	0.40268	0.59732	0.14701	3.5339	5.2421	3.5633	
53	Fe	0.59732	0.40268	0.14701	5.2421	3.5339	3.5633	
54	Fe	0.59813	0.40187	0.36057	5.2492	3.5268	8.7397	
55	Fe	0.40187	0.59813	0.36057	3.5268	5.2492	8.7397	
56	Fe	0.09681	0.09681	0.10257	0.8496	0.8496	2.4862	
57	Fe	0.70322	0.70322	0.37353	6.1715	6.1715	9.0539	
58	Fe	0.29678	0.29678	0.37353	2.6045	2.6045	9.0539	
59	Fe	0.68486	0.68486	0.12240	6.0103	6.0103	2.9668	
60	Fe	0.18508	0.81492	0.12707	1.6242	7.1518	3.0800	
61	Fe	0.81492	0.18508	0.12707	7.1518	1.6242	3.0800	
62	Fe	0.81310	0.18690	0.37166	7.1358	1.6402	9.0087	

#	El.	*x*	*y*	*z*	*X*	*Y*	*Z*	Site
63	Fe	0.18690	0.81310	0.37166	1.6402	7.1358	9.0087	
64	Fe	0.31514	0.31514	0.12240	2.7657	2.7657	2.9668	
65	Fe	-0.00000	-0.00000	0.30390	0.0000	0.0000	7.3663	
66	Fe	0.50000	0.50000	0.05696	4.3880	4.3880	1.3805	
67	Fe	0.50000	0.50000	0.44588	4.3880	4.3880	10.8077	
68	Fe	-0.00000	-0.00000	0.19182	0.0000	0.0000	4.6496	
69	Fe	0.50000	-0.00000	0.00097	4.3880	0.0000	0.0236	
70	Fe	0.50000	-0.00000	0.24982	4.3880	0.0000	6.0554	
71	Fe	-0.00000	0.50000	0.24982	-0.0000	4.3880	6.0554	
72	Fe	-0.00000	0.50000	0.00097	-0.0000	4.3880	0.0236	
73	Fe	0.77366	0.55952	0.56334	6.7897	4.9104	13.6548	
74	Fe	0.22634	0.44048	0.56334	1.9864	3.8656	13.6548	
75	Fe	0.52261	0.22916	0.94344	4.5865	2.0111	22.8678	
76	Fe	0.06083	0.72310	0.81070	0.5338	6.3459	19.6504	
77	Fe	0.93917	0.27690	0.81070	8.2422	2.4301	19.6504	
78	Fe	0.27574	0.93460	0.68595	2.4199	8.2021	16.6266	
79	Fe	0.72426	0.06540	0.68595	6.3562	0.5739	16.6266	
80	Fe	0.47739	0.77084	0.94344	4.1895	6.7649	22.8678	
81	Fe	0.22916	0.52261	0.94344	2.0111	4.5865	22.8678	
82	Fe	0.77084	0.47739	0.94344	6.7649	4.1895	22.8678	
83	Fe	0.55952	0.77366	0.56334	4.9104	6.7897	13.6548	
84	Fe	0.93460	0.27574	0.68595	8.2021	2.4199	16.6266	
85	Fe	0.06540	0.72426	0.68595	0.5739	6.3562	16.6266	
86	Fe	0.72310	0.06083	0.81070	6.3459	0.5338	19.6504	
87	Fe	0.27690	0.93917	0.81070	2.4301	8.2422	19.6504	
88	Fe	0.44048	0.22634	0.56334	3.8656	1.9864	13.6548	
89	Fe	0.96574	0.35905	0.58832	8.4753	3.1510	14.2602	
90	Fe	0.03426	0.64095	0.58832	0.3007	5.6250	14.2602	
91	Fe	0.64349	0.01341	0.91034	5.6473	0.1177	22.0657	
92	Fe	0.85556	0.53025	0.83880	7.5084	4.6535	20.3316	
93	Fe	0.14444	0.46975	0.83880	1.2676	4.1225	20.3316	
94	Fe	0.46475	0.14078	0.66169	4.0786	1.2355	16.0387	
95	Fe	0.53525	0.85922	0.66169	4.6974	7.5405	16.0387	
96	Fe	0.35651	0.98659	0.91034	3.1287	8.6583	22.0657	
97	Fe	0.01341	0.64349	0.91034	0.1177	5.6473	22.0657	
98	Fe	0.98659	0.35651	0.91034	8.6583	3.1287	22.0657	
99	Fe	0.35905	0.96574	0.58832	3.1510	8.4753	14.2602	
100	Fe	0.14078	0.46475	0.66169	1.2355	4.0786	16.0387	
101	Fe	0.85922	0.53525	0.66169	7.5405	4.6974	16.0387	
102	Fe	0.53025	0.85556	0.83880	4.6535	7.5084	20.3316	
103	Fe	0.46975	0.14444	0.83880	4.1225	1.2676	20.3316	
104	Fe	0.64095	0.03426	0.58832	5.6250	0.3007	14.2602	
105	Fe	0.09539	0.90461	0.88996	0.8372	7.9388	21.5716	
106	Fe	0.90461	0.09539	0.88996	7.9388	0.8372	21.5716	
107	Fe	0.09681	0.90319	0.60257	0.8496	7.9264	14.6057	
108	Fe	0.40268	0.40268	0.64701	3.5339	3.5339	15.6828	
109	Fe	0.59732	0.59732	0.64701	5.2421	5.2421	15.6828	

#	El.	x	y	z	X	Y	Z	Site
110	Fe	0.59813	0.59813	0.86057	5.2492	5.2492	20.8592	
111	Fe	0.40187	0.40187	0.86057	3.5268	3.5268	20.8592	
112	Fe	0.90319	0.09681	0.60257	7.9264	0.8496	14.6057	
113	Fe	0.29678	0.70322	0.87353	2.6045	6.1715	21.1733	
114	Fe	0.70322	0.29678	0.87353	6.1715	2.6045	21.1733	
115	Fe	0.31514	0.68486	0.62240	2.7657	6.0103	15.0862	
116	Fe	0.18508	0.18508	0.62707	1.6242	1.6242	15.1994	
117	Fe	0.81492	0.81492	0.62707	7.1518	7.1518	15.1994	
118	Fe	0.81310	0.81310	0.87166	7.1358	7.1358	21.1281	
119	Fe	0.18690	0.18690	0.87166	1.6402	1.6402	21.1281	
120	Fe	0.68486	0.31514	0.62240	6.0103	2.7657	15.0862	
121	Fe	-0.00000	-0.00000	0.80390	0.0000	0.0000	19.4857	
122	Fe	0.50000	0.50000	0.55696	4.3880	4.3880	13.5000	
123	Fe	0.50000	0.50000	0.94588	4.3880	4.3880	22.9271	
124	Fe	-0.00000	-0.00000	0.69182	0.0000	0.0000	16.7690	
125	Fe	-0.00000	0.50000	0.50097	-0.0000	4.3880	12.1430	
126	Fe	-0.00000	0.50000	0.74982	-0.0000	4.3880	18.1748	
127	Fe	0.50000	-0.00000	0.74982	4.3880	0.0000	18.1748	
128	Fe	0.50000	-0.00000	0.50097	4.3880	0.0000	12.1430	
129	B	0.62371	0.37629	0.00056	5.4737	3.3023	0.0135	
130	B	0.12540	0.12540	0.24847	1.1005	1.1005	6.0226	
131	B	0.87460	0.87460	0.24847	7.6755	7.6755	6.0226	
132	B	0.37629	0.62371	0.00056	3.3023	5.4737	0.0135	
133	B	0.62371	0.62371	0.50056	5.4737	5.4737	12.1330	
134	B	0.87460	0.12540	0.74847	7.6755	1.1005	18.1420	
135	B	0.12540	0.87460	0.74847	1.1005	7.6755	18.1420	
136	B	0.37629	0.37629	0.50056	3.3023	3.3023	12.1330	

Table S3: Coordination of every rare-earth site: number of Fe neighbors within 3.3 Å and the eight shortest RE–Fe distances (periodic images included).

Environment	Atom	Site	z (Å)	N_{Fe}	shortest RE–Fe distances (Å)
bulk	1	4g	0.000	14	3.056, 3.056, 3.056, 3.056, 3.123, 3.123, 3.265, 3.265
bulk	2	4g	6.065	14	3.056, 3.056, 3.056, 3.056, 3.123, 3.123, 3.265, 3.265
bulk	3	4g	6.065	14	3.056, 3.056, 3.056, 3.056, 3.123, 3.123, 3.265, 3.265
bulk	4	4g	0.000	14	3.056, 3.056, 3.056, 3.056, 3.123, 3.123, 3.265, 3.265
bulk	5	4f	0.000	16	3.042, 3.042, 3.054, 3.054, 3.054, 3.054, 3.064, 3.064
bulk	6	4f	6.065	16	3.042, 3.042, 3.054, 3.054, 3.054, 3.054, 3.064, 3.064
bulk	7	4f	6.065	16	3.042, 3.042, 3.054, 3.054, 3.054, 3.054, 3.064, 3.064
bulk	8	4f	0.000	16	3.042, 3.042, 3.054, 3.054, 3.054, 3.054, 3.064, 3.064
GB	1	class B	0.052	10	2.817, 2.982, 2.982, 3.051, 3.103, 3.103, 3.214, 3.214
GB	2	class C	6.065	10	3.057, 3.057, 3.071, 3.071, 3.078, 3.140, 3.224, 3.224
GB	3	class C	6.065	10	3.057, 3.057, 3.071, 3.071, 3.078, 3.140, 3.224, 3.224
GB	4	class B	0.052	10	2.817, 2.982, 2.982, 3.051, 3.103, 3.103, 3.214, 3.214
GB	5	class A	-0.035	13	2.766, 2.766, 3.032, 3.032, 3.104, 3.104, 3.118, 3.138
GB	6	class D	6.061	15	3.009, 3.010, 3.010, 3.046, 3.054, 3.054, 3.081, 3.081

Environment	Atom	Site	z (Å)	N_{Fe}	shortest RE–Fe distances (Å)
GB	7	class D	6.061	15	3.009, 3.010, 3.010, 3.046, 3.054, 3.054, 3.081, 3.081
GB	8	class A	-0.035	13	2.766, 2.766, 3.032, 3.032, 3.104, 3.104, 3.118, 3.138
GB	9	class B	12.172	10	2.817, 2.982, 2.982, 3.051, 3.103, 3.103, 3.214, 3.214
GB	10	class C	18.185	10	3.057, 3.057, 3.071, 3.071, 3.078, 3.140, 3.224, 3.224
GB	11	class C	18.185	10	3.057, 3.057, 3.071, 3.071, 3.078, 3.140, 3.224, 3.224
GB	12	class B	12.172	10	2.817, 2.982, 2.982, 3.051, 3.103, 3.103, 3.214, 3.214
GB	13	class A	12.084	13	2.766, 2.766, 3.032, 3.032, 3.104, 3.104, 3.118, 3.138
GB	14	class D	18.180	15	3.009, 3.010, 3.010, 3.046, 3.054, 3.054, 3.081, 3.081
GB	15	class D	18.180	15	3.009, 3.010, 3.010, 3.046, 3.054, 3.054, 3.081, 3.081
GB	16	class A	12.084	13	2.766, 2.766, 3.032, 3.032, 3.104, 3.104, 3.118, 3.138

Table S4: Many-electron QDPT-SOC descriptors of the finite-cell calculations (cm^{-1}). $\delta_{01} = E_1 - E_0$ is zero for the Kramers ions; for Pr and Tb it is the lowest-pair splitting. $E_{\mathrm{next}} = E_2 - E_0$ is the next Kramers doublet or the level above the lowest non-Kramers pair. W_J is the span of the assigned ground-J manifold. These quantities are distinct from the AILFT orbital gaps in main Table 1.

Ion	δ_{01}^{b}	δ_{01}^{g}	E_{next}^{b}	E_{next}^{g}	W_J^{b}	W_J^{g}
Ce	0.00	0.00	1506.36	829.43	2040.24	1420.66
Pr	5.66	0.22	308.32	470.19	1053.86	1193.95
Nd	0.00	0.00	80.27	162.04	371.26	447.54
Gd	0.00	0.00	1.14	1.23	2.32	2.72
Tb	0.00	0.00	266.74	284.24	1061.73	1290.15
Dy	0.00	0.00	138.54	247.27	917.43	1172.93

Table S5: Exact axial parameter and rank invariants of the finite-cell AILFT operator (cm^{-1}). Bulk = atom 2, interface = atom 13. The full complex Wybourne sets for these and the other archived calculations accompany the PDF as `wybourne_Bkq_all_runs.csv`.

Ion	Site	$A_2^0\langle r^2\rangle$	S_2	S_4	S_6
Ce	bulk	1730.19	1673.98	606.69	514.05
Ce	GB	-1023.40	1223.48	46.75	6.18
Pr	bulk	1403.31	1283.71	364.69	211.87
Pr	GB	-985.09	1179.16	43.85	5.57
Nd	bulk	1160.69	1049.63	235.06	97.28
Nd	GB	-932.98	1118.28	39.43	4.65
Gd	bulk	770.04	768.88	88.98	21.31
Gd	GB	-808.55	970.20	27.25	2.80
Tb	bulk	752.05	752.69	83.17	19.87
Tb	GB	-797.63	956.51	25.57	2.47
Dy	bulk	716.11	727.20	72.29	15.64
Dy	GB	-776.92	931.61	23.52	2.25

Table S6: Selected sensitivity calculations at fixed centers (cm^{-1}). Sphere models use Fe $-3/14$, B -3 unless marked alternative (Fe $-5/14$, B -1). All use DKH2/SARC2-DKH-QZVP. NEVPT2 rows report the corrected block, not its accompanying CASSCF block. E_{next} is the first excited Kramers doublet for these two ions. The finite-cell versus sphere step changes multiple embedding choices; only the 18/25 Å sphere comparison isolates radius. These two centers are not a matched parent-sublattice pair.

Ion	Center	Model	Orbital spread	$A_2^0\langle r^2\rangle$	E_{next}
Nd	bulk 2	finite cell	1463.8	1160.7	80.27
Nd	bulk 2	sphere 18 A	1490.5	1183.6	103.45
Nd	bulk 2	sphere 25 A	1411.6	1118.7	95.44
Nd	bulk 2	alternative charges	1823.1	1492.0	169.63
Nd	bulk 2	finite cell + NEVPT2	1277.3	1016.2	80.93
Nd	bulk 2	sphere 25 A + NEVPT2	1220.7	970.8	88.76
Nd	GB 13	finite cell	1513.2	-933.0	162.04
Nd	GB 13	sphere 18 A	1383.9	731.2	51.44
Nd	GB 13	sphere 25 A	1510.4	784.0	55.00
Nd	GB 13	alternative charges	1871.8	1135.8	100.44
Nd	GB 13	finite cell + NEVPT2	1302.2	-804.4	145.66
Nd	GB 13	sphere 25 A + NEVPT2	1345.1	693.5	79.35
Dy	bulk 2	finite cell	1001.6	716.1	138.54
Dy	bulk 2	sphere 18 A	1097.7	814.7	164.82
Dy	bulk 2	sphere 25 A	1038.7	765.0	153.00
Dy	bulk 2	alternative charges	1378.4	1147.9	258.14
Dy	bulk 2	finite cell + NEVPT2	823.3	591.3	121.97
Dy	bulk 2	sphere 25 A + NEVPT2	854.2	630.6	129.02
Dy	GB 13	finite cell	1261.4	-776.9	247.27
Dy	GB 13	sphere 18 A	750.5	447.8	64.80
Dy	GB 13	sphere 25 A	802.2	476.6	68.84
Dy	GB 13	alternative charges	1340.2	846.3	153.79
Dy	GB 13	finite cell + NEVPT2	1037.1	-638.4	224.01
Dy	GB 13	sphere 25 A + NEVPT2	670.1	397.7	32.53

Table S7: Matched pairs in the 25 Å sphere model (energies in cm^{-1}). A: bulk $4f$ atom 5 $\rightarrow$ interface class A atom 13. B: bulk $4g$ atom 2 $\rightarrow$ class B atom 9. E_{next} is defined as in Table S4. The final column is the interface axis tilt from c in degrees: direct ground-doublet magnetic matrix elements for Ce/Nd/Dy. Pr, Tb and Gd directions are omitted (Section S4). Ce class A is unavailable.

Ion	Pair	Orbital spread (b→g)	$A_2^0\langle r^2\rangle$ (b→g)	E_{next} (b→g)	Tilt
Pr	A	1869.6→2143.1	742.3→968.4	63.69→120.49	–
Nd	A	1251.3→1510.4	568.8→784.0	32.93→55.00	1.2
Gd	A	637.2→867.2	331.1→512.7	0.28→0.46	–
Tb	A	619.7→847.8	321.6→500.8	80.86→148.99	–
Dy	A	578.6→802.2	302.5→476.6	32.19→68.84	0.6
Ce	B	1935.8→2007.4	1501.4→1563.2	1273.14→1311.85	4.9
Pr	B	1629.0→1733.1	1295.3→1330.5	316.92→309.22	–
Nd	B	1411.6→1549.9	1118.7→1135.4	95.44→81.38	13.7
Gd	B	1097.7→1246.9	814.4→805.4	1.10→1.10	–
Tb	B	1075.8→1222.6	797.3→787.3	266.82→254.33	–
Dy	B	1038.7→1182.7	765.0→753.7	153.00→145.73	12.8

Table S8: Capped effective-core-potential (cECP) test (cm^{-1}). All Fe and rare-earth neighbors within 4.0 Å of the ion (18–19 centers) are replaced by capped ECPs with the same formal charges (Hay–Wadt for Fe, def2 small-core for the rare earth); pc = the same calculation with bare point charges. Level: def2-TZVP with the def2 small-core ECP on the central ion, spin-free SA-CASSCF + AILFT, $R = 25$ Å. The last column is the sphere-model DKH2/SARC2-DKH-QZVP value with bare point charges.

Ion	Site	split pc	split cECP	$A_2^0\langle r^2\rangle$ pc	$A_2^0\langle r^2\rangle$ cECP	S_6 pc	S_6 cECP	$A_2^0\langle r^2\rangle$ DKH
Nd	bulk $4f$ (5)	1225	1195	569	561	86	79	569
	GB A (13)	1488	1459	788	781	93	86	784
	bulk $4g$ (2)	1425	1421	1119	1116	69	66	1119
	GB B (9)	1566	1564	1130	1128	54	52	1135
Dy	bulk $4f$ (5)	494	490	267	266	12	12	302
	GB A (13)	711	708	435	434	14	14	477
	bulk $4g$ (2)	1001	1000	714	713	13	13	765
	GB B (9)	1148	1147	698	697	9	9	754

Table S9: Conditional Stoner–Wohlfarth loss-of-metastability field for the finite-cell susceptibility-axis angles, assuming the applied field is along the bulk response axis. The listed value is dimensionless; it assumes an isolated uniaxial region with the same K and M_s at each angle. It is not a prediction of nucleation or macroscopic coercivity.

Ion	θ (degrees)	H_{sw}/H_A
Ce	87.00	0.823
Pr	87.35	0.835
Nd	87.64	0.845
Tb	87.27	0.832
Dy	87.45	0.838

Table S10: Full-cube integrals for the individual AILFT orbital in main Fig. 5. N is the numerical one-electron norm before normalization; r_{rms} uses the full three-dimensional density and ion-centered distances. Cubes have 80^3 points with spacing about 0.094 Å. These are eigenorbital metrics of the finite-cell models, not total correlated-shell radii.

Ion	Environment	Grid norm N	r_{rms} (Å)
Ce	Bulk	0.99841	0.8374
Ce	GB	1.00181	0.6095
Pr	Bulk	0.99894	0.6612
Pr	GB	1.00232	0.5849
Dy	Bulk	1.00035	0.4715
Dy	GB	1.00878	0.4646